

Department of Electrical Engineering
Dr. B. C. Roy Engineering College, Durgapur

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Complete Class Notes

The Exponential Family

“Mathematics is the language in which God has written the universe.”
— Galileo Galilei

 **Subject:** Engineering Mathematics | Control Systems | Signal Processing
 **Prepared by:** Kingsuk Majumdar, Ph.D. (Engg.)
 **Institute:** BCREC Durgapur, Dept. of Electrical Engineering
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 **Contact:** kingsuk.majumdar@bcrec.ac.in

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1. Foundational Concepts: What is e ?

1.1. The Definition of e

Euler's number e is defined in multiple equivalent ways:

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = 2.71828182845 \dots \quad (1)$$

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots \quad (2)$$

1.2. Why is e Special? (The Self-Derivative Property)

The Defining Property

$$\frac{d}{dx}(e^x) = e^x \quad \Leftarrow \quad \text{The ONLY function that is its own derivative!} \quad (3)$$

Table 1. Properties of e^x arising from self-derivative

Property	Mathematical Statement
Derivative	$\frac{d}{dx}(e^x) = e^x$
Integral	$\int e^x dx = e^x + C$
n th Derivative	$\frac{d^n}{dx^n}(e^x) = e^x$ for all n
Value at origin	$e^0 = 1$

Key Insight

A system governed by e^x grows at a rate exactly equal to its current size. This is the mathematical description of **compounding** – whether it is compound interest, population growth, or nuclear chain reactions.

1.3. The Taylor / Maclaurin Series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad (4)$$

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots \quad (5)$$

First-order engineering approximation

$$e^x \approx 1 + x \quad \text{valid when } |x| \ll 1$$

Used extensively in linearisation of nonlinear systems around an operating point.

1.4. Key Identities to Memorise

$$e^{a+b} = e^a \cdot e^b \quad e^{a-b} = \frac{e^a}{e^b} \quad (e^a)^b = e^{ab} \quad (6)$$

$$e^0 = 1 \quad e^1 = e \approx 2.718 \quad e^{-1} \approx 0.368 \quad \boxed{e^{j\pi} = -1} \quad (7)$$

Classroom Analogy

Euler's Identity $e^{j\pi} + 1 = 0$ is considered the most beautiful equation in mathematics, connecting e , π , j , 1, and 0 in a single compact statement.

2. The Pure Growth Function: $f(x) = e^x$

2.1. Mathematical Properties

Table 2. Properties of $f(x) = e^x$

Property	Value
Domain	$(-\infty, \infty)$
Range	$(0, \infty)$
Y-Intercept	$(0, 1)$
Horizontal Asymptote	HA: $y = 0$ as $x \rightarrow -\infty$
Vertical Asymptote	None
Derivative	e^x (self-referential)
Integral	$e^x + C$
Convexity	Always concave up

2.2. Conceptual Derivation

If a quantity Q grows proportionally to its current value:

$$\frac{dQ}{dt} = kQ \implies Q(t) = Q_0 e^{kt} \quad (8)$$

This single ODE models bacterial population, radioactive decay (negative k), compound interest, and amplifier gain.

2.3. Engineering Application: Thermal Runaway in BJT

In a Bipolar Junction Transistor (BJT):

$$I_C = I_S e^{V_{BE}/V_T} \quad (9)$$

where $V_T = kT/q \approx 26 \text{ mV}$ at room temperature and I_S is the reverse saturation current. A temperature rise increases I_C exponentially, generating more heat – a positive feedback loop called **thermal runaway**.

2.4. Figure

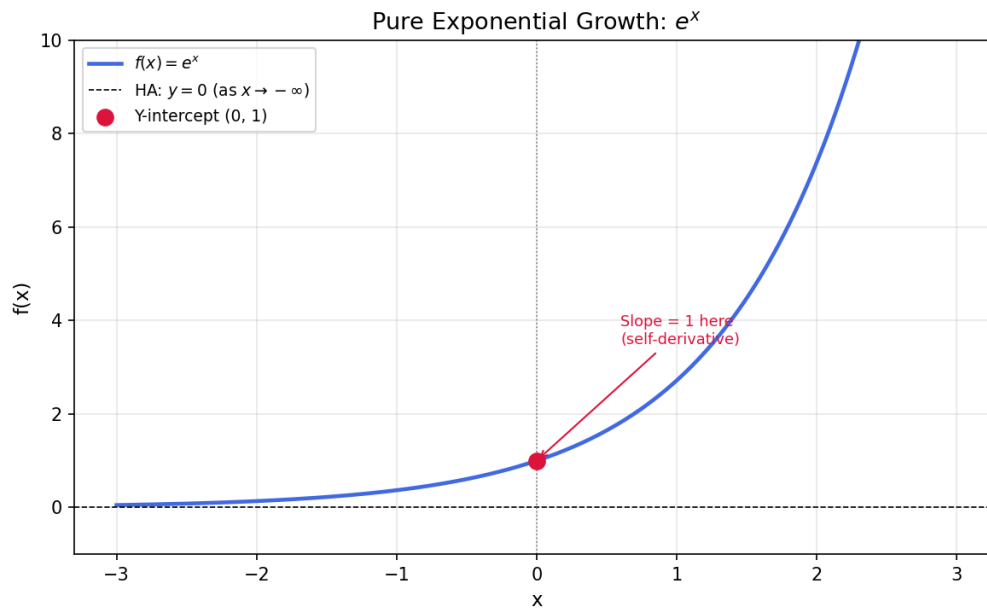


Figure 1. Pure Exponential Growth: $f(x) = e^x$. Note the self-derivative property annotated at $x = 0$.

2.5. Python Code

```

1  #!/usr/bin/env python3
2  # -*- coding: utf-8 -*-
3  # Title   : Pure Exponential Growth -  $e^x$ 
4  # Author  : Kingsuk Majumdar
5  # Version : V1 | 2026-07-13
6  # "Growth unchecked is the mathematics of catastrophe."
7
8  import numpy as np
9  import matplotlib.pyplot as plt
10 import os
11
12 script_dir = os.path.dirname(os.path.abspath(__file__))
13 output_dir = os.path.join(script_dir, 'OUTPUT')
14 os.makedirs(output_dir, exist_ok=True)
15
16 x = np.linspace(-3, 3, 500)
17 y = np.exp(x)
18
19 fig, ax = plt.subplots(figsize=(8, 5))
20 ax.plot(x, y, label=r'$f(x) = e^x$', color='royalblue', linewidth=2.2)
21 ax.axhline(0, color='black', lw=0.8, linestyle='--',
22           label=r'HA: $y=0$ (as $x \to -\infty$)')
23 ax.scatter(0, 1, color='crimson', zorder=5, s=80, label='Y-intercept (0, 1)')
24 ax.set_ylim(-1, 10)
25 ax.set_xlabel('x'); ax.set_ylabel('f(x)')
26 ax.set_title(r'Pure Exponential Growth:  $e^x$ ')
27 ax.legend(); ax.grid(True)
28 fig.tight_layout()
29 fig.savefig(os.path.join(output_dir, 'S01_fig01_exp_growth.png'), dpi=150)

```

Listing 1. Plotting e^x

3. The Pure Decay Function: $f(x) = e^{-x}$

3.1. Mathematical Properties

Table 3. Properties of $f(x) = e^{-x}$

Property	Value
Domain	$(-\infty, \infty)$
Range	$(0, \infty)$
Y-Intercept	$(0, 1)$
Horizontal Asymptote	HA: $y = 0$ as $x \rightarrow +\infty$
Derivative	$-e^{-x}$ (always decreasing)
Convexity	Always concave up

3.2. Engineering Application: RC Discharging Circuit

$$V(t) = V_0 e^{-t/RC} = V_0 e^{-t/\tau}, \quad \tau = RC \quad (10)$$

Table 4. The 5- τ rule for RC discharge

Time elapsed	Voltage remaining
$t = \tau$	36.8% of V_0
$t = 2\tau$	13.5% of V_0
$t = 3\tau$	5.0% of V_0
$t = 5\tau$	0.67% of V_0 (practically zero)

Rule of thumb in power electronics

A circuit is considered “fully discharged” after 5τ . The orange markers on Fig. 2 show these checkpoints.

3.3. Figure

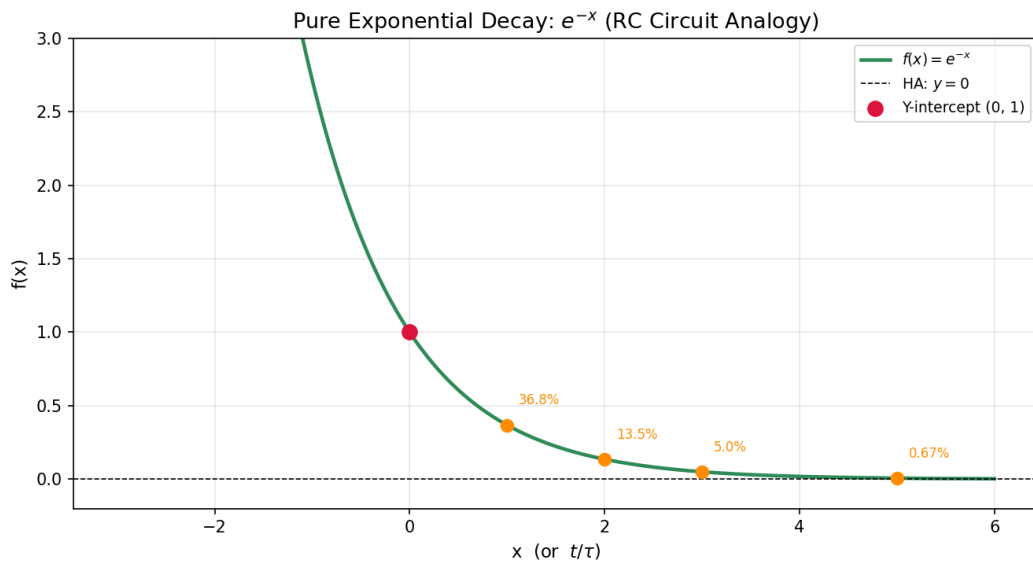


Figure 2. Pure Exponential Decay: $f(x) = e^{-x}$ with the $5\text{-}\tau$ checkpoints annotated (RC circuit analogy).

3.4. Python Code

```

1 x = np.linspace(-3, 6, 500)
2 y = np.exp(-x)
3 tau_points = [1, 2, 3, 5]
4 pct        = [36.8, 13.5, 5.0, 0.67]
5
6 fig, ax = plt.subplots(figsize=(9, 5))
7 ax.plot(x, y, color='seagreen', label=r'$f(x) = e^{-x}$')
8 for t, p in zip(tau_points, pct):
9     ax.scatter(t, np.exp(-t), color='darkorange', zorder=5, s=55)
10    ax.annotate(f'{p}%', xy=(t, np.exp(-t)),
11               xytext=(t+0.12, np.exp(-t)+0.14), fontsize=8)
12 fig.savefig(os.path.join(output_dir, 'S01_fig02_exp_decay.png'), dpi=150)

```

Listing 2. Plotting e^{-x} with $5\text{-}\tau$ markers

4. The Saturating Rise: $f(x) = 1 - e^{-x}$

4.1. Mathematical Properties

Table 5. Properties of $f(x) = 1 - e^{-x}$

Property	Value
Domain	$(-\infty, \infty)$
Range	$(-\infty, 1)$
X and Y-Intercept	$(0, 0)$
Horizontal Asymptote	HA: $y = 1$ as $x \rightarrow +\infty$
Derivative	e^{-x} (always positive – always increasing)
Convexity	Concave down for $x > 0$

4.2. Derivation from the ODE

The charging RC circuit satisfies:

$$\frac{dV}{dt} = \frac{V_{\max} - V}{\tau} \Rightarrow V(t) = V_{\max}(1 - e^{-t/\tau}) \quad (11)$$

The driving force is proportional to the **remaining gap** ($V_{\max} - V$). As the gap shrinks, so does the rate of change – hence the saturating shape.

4.3. Engineering Application: RC Charging / First-Order Step Response

Used in: step response of a first-order control plant, current buildup in RL circuits, and ON-delay timer output in PLC ladder logic.

Key Insight

At $t = 1\tau$, the output reaches $1 - e^{-1} \approx 63.2\%$ of its final value. This is the single most tested result in GATE and university exams on control systems.

4.4. Figure

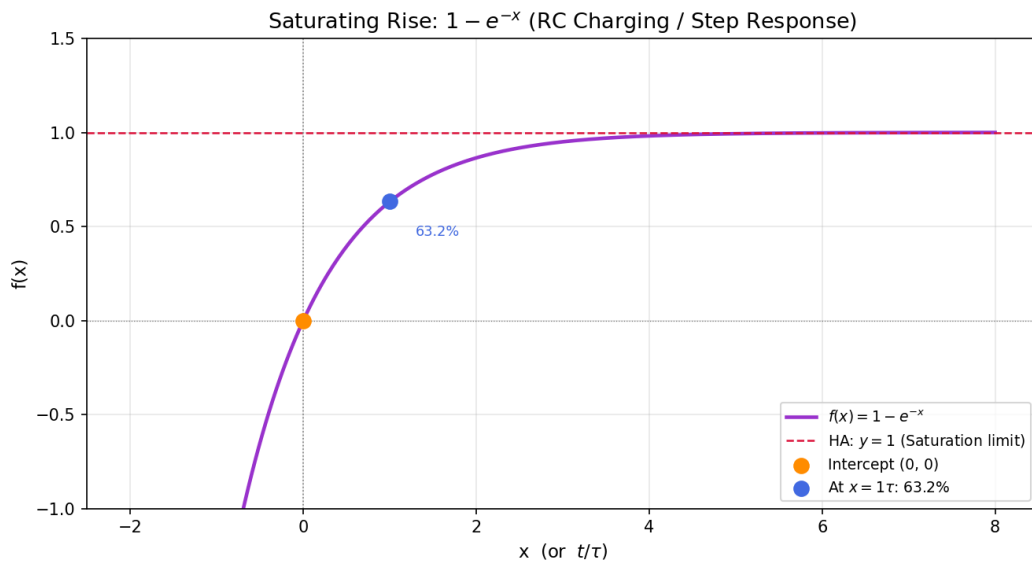


Figure 3. Saturating Rise: $f(x) = 1 - e^{-x}$ – RC charging / step response. The 63.2% point at $x = 1\tau$ is marked in blue.

4.5. Python Code

```

1 x = np.linspace(-2, 8, 500)
2 y = 1 - np.exp(-x)
3
4 ax.plot(x, y, color='darkorchid', label=r'$f(x) = 1 - e^{-x}$')
5 ax.axhline(1, color='crimson', lw=1.2, linestyle='--',
6           label=r'HA: $y = 1$ (Saturation limit)')
7 ax.scatter(1, 1 - np.exp(-1), color='royalblue', zorder=5, s=80,
8           label=r'At $x = 1\tau$: 63.2%')

```

Listing 3. Plotting $1 - e^{-x}$ with 63.2% annotation

5. The Saturating Drop: $f(x) = 1 - e^x$

5.1. Mathematical Properties

Table 6. Properties of $f(x) = 1 - e^x$

Property	Value
Domain	$(-\infty, \infty)$
Range	$(-\infty, 1)$
Intercepts	$(0, 0)$
Horizontal Asymptote	HA: $y = 1$ as $x \rightarrow -\infty$
Derivative	$-e^x$ (always negative – always decreasing)

5.2. Relationship to Section 3

$1 - e^x$ is the mirror image of $1 - e^{-x}$ about the Y-axis. If time runs “backwards”, the charging curve becomes the saturating drop.

5.3. Engineering Application: Electromagnetic Braking

When an inductor or electromagnet is suddenly short-circuited, the braking force drops in a pattern described by this function – ensuring a predictable and safe deceleration profile in fail-safe braking systems.

5.4. Figure

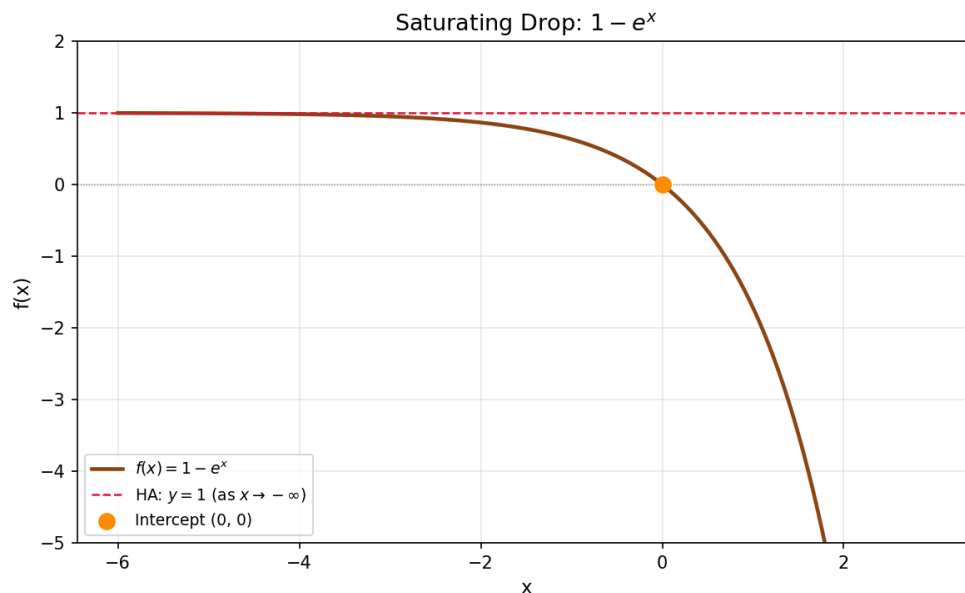


Figure 4. Saturating Drop: $f(x) = 1 - e^x$ – mirror image of the charging curve.

6. The Negative Decay: $f(x) = -e^{-x}$

6.1. Mathematical Properties

Table 7. Properties of $f(x) = -e^{-x}$

Property	Value
Domain	$(-\infty, \infty)$
Range	$(-\infty, 0)$
Y-Intercept	$(0, -1)$
Horizontal Asymptote	HA: $y = 0$ from <i>below</i> as $x \rightarrow +\infty$
Derivative	e^{-x} (positive – function increases toward 0)

Key Insight

The function is always **negative** and always **increasing** (becoming less negative). The asymptote $y = 0$ is approached from the negative side.

6.2. Engineering Application: Inverting Op-Amp Integrator

For an inverting configuration, the output is always negative, decaying toward zero:

$$V_{\text{out}}(t) \approx -V_0 e^{-t/RC} \quad (12)$$

6.3. Figure

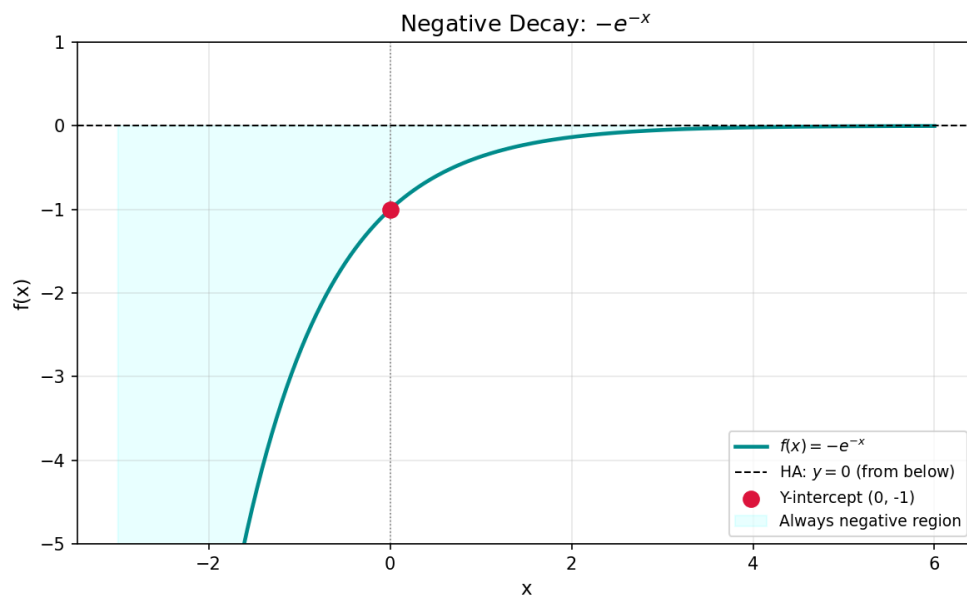


Figure 5. Negative Decay: $f(x) = -e^{-x}$ – always negative, approaching zero from below. The shaded cyan region highlights the always-negative domain.

7. The Standard Sigmoid (Logistic Function): $f(x) = 1/(1 + e^{-x})$

7.1. Mathematical Properties

Table 8. Properties of the Sigmoid $\sigma(x)$

Property	Value
Domain	$(-\infty, \infty)$
Range	$(0, 1)$
Symmetry	Rotational about $(0, 0.5)$
Intercept	$(0, 0.5)$
HA (left)	HA: $y = 0$ as $x \rightarrow -\infty$
HA (right)	HA: $y = 1$ as $x \rightarrow +\infty$
Max Slope	0.25 at $x = 0$

7.2. Derivation from Logistic ODE

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right) \Rightarrow P(t) = \frac{K}{1 + e^{-r(t-t_0)}} \quad (13)$$

7.3. Derivative of Sigmoid

Self-referential derivative – critical for backpropagation

$$\sigma'(x) = \sigma(x)(1 - \sigma(x)) \quad (14)$$

Also note: $\sigma(x) + \sigma(-x) = 1$ always (odd symmetry about the centre).

7.4. Engineering Application: Neural Networks and Soft-Starting

As an activation function: Output = $\sigma(\sum w_i x_i + b)$ squashes any real input into $(0, 1)$. In power drives, a sigmoid-shaped speed ramp limits inrush current during motor startup.

7.5. Figure

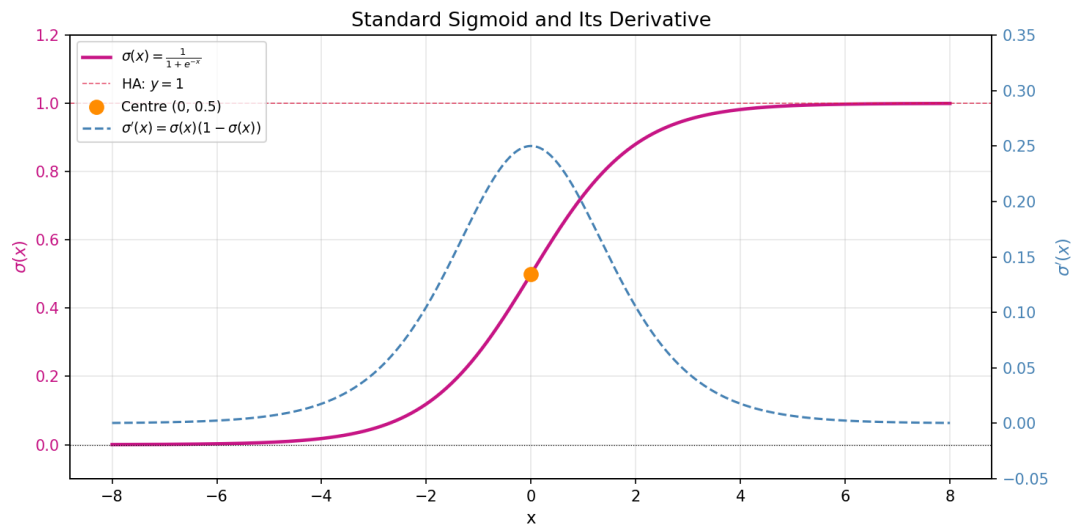


Figure 6. Standard Sigmoid $\sigma(x)$ (left axis, magenta) and its derivative $\sigma'(x) = \sigma(x)(1 - \sigma(x))$ (right axis, blue dashed).

7.6. Python Code

```

1 x = np.linspace(-8, 8, 500)
2 sigma = 1 / (1 + np.exp(-x))
3 sigma_deriv = sigma * (1 - sigma)
4
5 fig, ax1 = plt.subplots(figsize=(10, 5))
6 ax2 = ax1.twinx()
7 ax1.plot(x, sigma, color='mediumvioletred',
8         label=r'$\sigma(x) = \frac{1}{1+e^{-x}}$')
9 ax2.plot(x, sigma_deriv, color='steelblue', lw=1.5, linestyle='--',
10        label=r"$\sigma'(x) = \sigma(x)(1-\sigma(x))$")

```

Listing 4. Sigmoid and its derivative – twin Y-axis plot

8. The Hyperbolic Tangent: $f(x) = \tanh(x)$

8.1. Mathematical Properties

Table 9. Properties of $\tanh(x)$

Property	Value
Domain	$(-\infty, \infty)$
Range	$(-1, 1)$
Symmetry	Odd function: $\tanh(-x) = -\tanh(x)$
Intercept	$(0, 0)$
HA (left)	HA: $y = -1$ as $x \rightarrow -\infty$
HA (right)	HA: $y = +1$ as $x \rightarrow +\infty$
Derivative	$1 - \tanh^2(x) = \operatorname{sech}^2(x)$
Max slope at origin	1

8.2. Relationship Between $\tanh$ and Sigmoid

$$\tanh(x) = 2\sigma(2x) - 1 \quad \sigma(x) = \frac{\tanh(x/2) + 1}{2} \quad (15)$$

8.3. Why $\tanh$ is Preferred over Sigmoid in Hidden Layers

Table 10. Sigmoid vs $\tanh$ comparison

Feature	Sigmoid	$\tanh$
Output range	$(0, 1)$	$(-1, 1)$
Zero-centred	No	Yes
Gradient at centre	0.25	1.0
Vanishing gradient	Worse	Better

Key Insight

Zero-centred outputs mean gradients during backpropagation are more balanced, leading to faster convergence in deep networks.

8.4. Engineering Application: Transmission Lines and LSTM Networks

Voltage along a lossy transmission line:

$$V(x) = V^+ \cosh(\gamma x) + V^- \sinh(\gamma x) \quad (16)$$

In LSTM networks for load forecasting or PV output prediction, $\tanh$ is used as the cell-state activation, keeping outputs bounded in $(-1, 1)$.

8.5. Figure

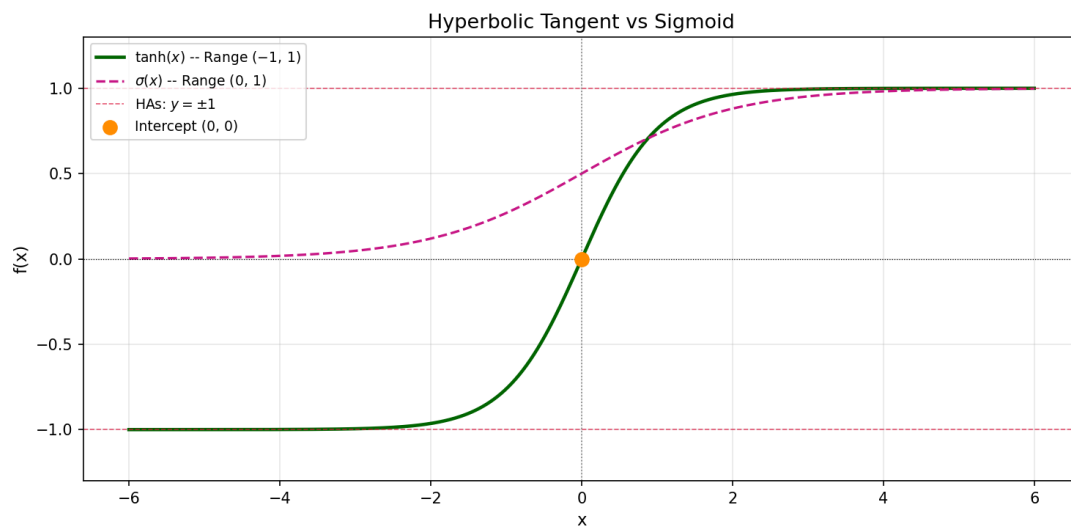


Figure 7. $\tanh(x)$ (green) vs Sigmoid $\sigma(x)$ (magenta dashed). Both asymptotes at $y = \pm 1$ are marked in red.

9. The Catenary / U-Shape: $f(x) = e^x + e^{-x} = 2 \cosh(x)$

9.1. Mathematical Properties

Table 11. Properties of $f(x) = e^x + e^{-x}$

Property	Value
Domain	$(-\infty, \infty)$
Range	$[2, \infty)$
Symmetry	Even function: $f(-x) = f(x)$
Minimum	$(0, 2)$
Asymptotes	None (both tails go to $+\infty$)
Derivative	$e^x - e^{-x} = 2 \sinh(x)$

9.2. The Catenary: A Natural Minimum Energy Shape

A flexible inextensible cable hanging between two supports takes the shape of a **catenary** – not a parabola (a common misconception).

$$y = a \cosh\left(\frac{x}{a}\right) \quad (17)$$

where a is the catenary parameter related to cable tension and weight per unit length.

9.3. Engineering Application: High-Voltage Transmission Line Sag

The sag d in a horizontal conductor span L :

$$d \approx \frac{wL^2}{8T} \quad (\text{parabolic approximation for } d/L < 0.1) \quad (18)$$

The exact shape follows cosh. Power system engineers calculate precise sag for tower design and ground clearance compliance.

9.4. Figure

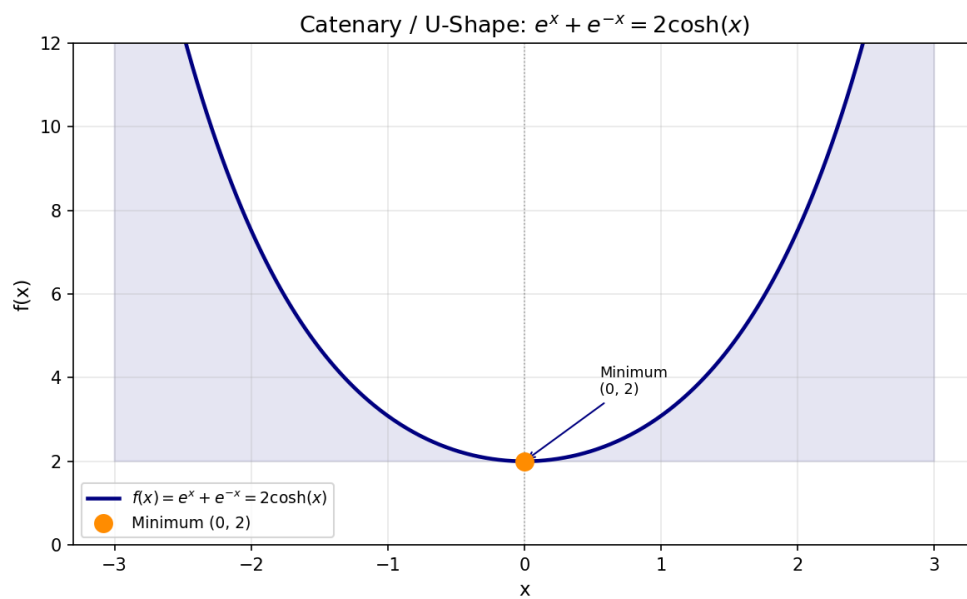


Figure 8. Catenary / U-Shape: $f(x) = e^x + e^{-x} = 2\cosh(x)$. Minimum at $(0, 2)$; shaded region highlights the bowl above the minimum.

10. The Reciprocal Sigmoid: $f(x) = 1/(1 - e^{-x})$ – Vertical Asymptote

10.1. Mathematical Properties

Table 12. Properties of $f(x) = 1/(1 - e^{-x})$

Property	Value
Domain	$\mathbb{R} \setminus \{0\}$ – all reals except $x = 0$
Range	$(-\infty, 0) \cup (1, \infty)$
Intercepts	None (undefined at $x = 0$)
Vertical Asymptote	VA: $x = 0$
HA (left)	HA: $y = 0^-$ as $x \rightarrow -\infty$
HA (right)	HA: $y = 1^+$ as $x \rightarrow +\infty$

Warning

This function is **DISCONTINUOUS** at $x = 0$. Never attempt to evaluate or plot it at zero without splitting the domain into $x < 0$ and $x > 0$.

10.2. Why the Vertical Asymptote Occurs at $x = 0$

The denominator $1 - e^{-x}$ equals zero when:

$$1 - e^{-x} = 0 \implies e^{-x} = 1 \implies -x = 0 \implies \boxed{x = 0} \quad (19)$$

10.3. Behaviour Around the Vertical Asymptote

Table 13. One-sided limits at $x = 0$

Approach direction	Limit
From the right ($x \rightarrow 0^+$)	$f(x) \rightarrow +\infty$ (shoots up)
From the left ($x \rightarrow 0^-$)	$f(x) \rightarrow -\infty$ (plummets down)

10.4. Engineering Application: Mechanical Resonance

Undamped amplitude of a forced spring-mass system:

$$A(\omega) = \frac{F_0/m}{\omega_n^2 - \omega^2} \rightarrow \infty \quad \text{as } \omega \rightarrow \omega_n \quad (20)$$

Real systems have damping ($\zeta > 0$) that limits the peak, but undamped models show the vertical asymptote clearly. Historical example: **Tacoma Narrows Bridge (1940)** – wind-induced resonance caused catastrophic structural collapse.

10.5. Figure

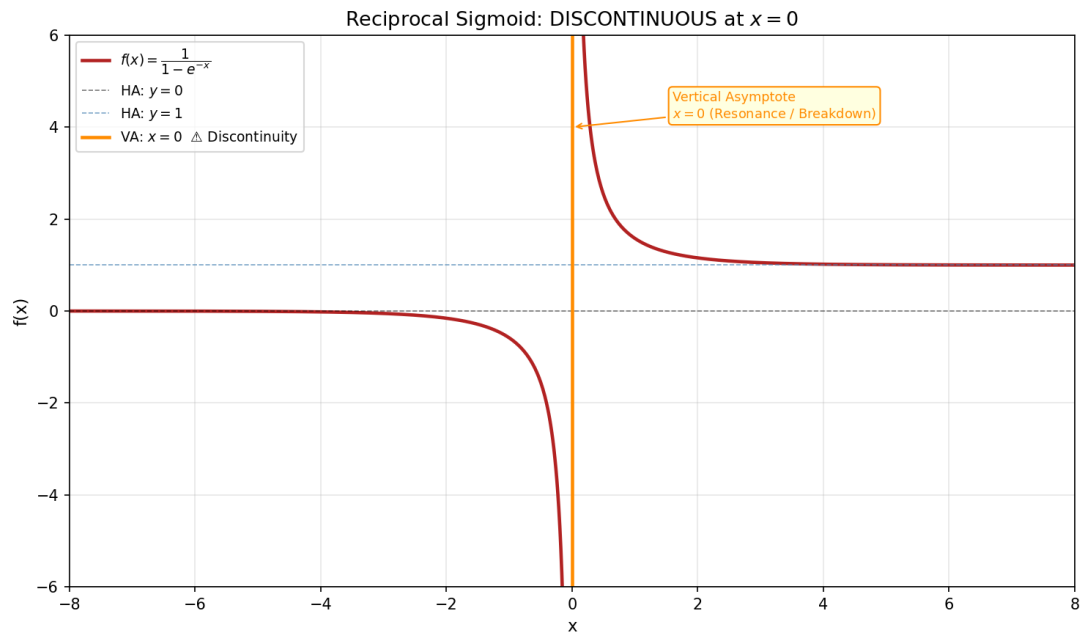


Figure 9. Reciprocal Sigmoid: DISCONTINUOUS at $x = 0$. The orange vertical line marks the asymptote; left and right branches are plotted separately.

10.6. Python Code

```

1 x_left = np.linspace(-8, -0.05, 500)
2 x_right = np.linspace( 0.05,  8, 500)
3
4 def rec_sig(x_arr):
5     return 1.0 / (1.0 - np.exp(-x_arr))
6
7 ax.plot(x_left, rec_sig(x_left), color='firebrick')
8 ax.plot(x_right, rec_sig(x_right), color='firebrick')
9 ax.axvline(0, color='darkorange', lw=2.2, label='VA: x = 0')
```

Listing 5. Splitting domain to handle the vertical asymptote

11. Euler Forms and Complex Exponentials

11.1. Euler's Formula

The Bridge Between Exponentials and Sinusoids

$$e^{j\theta} = \cos \theta + j \sin \theta \quad e^{-j\theta} = \cos \theta - j \sin \theta \quad (21)$$

11.2. Recovering Trig Functions from Euler

$$\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2} \quad \sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j} \quad (22)$$

11.3. Pure Rotation: $e^{j\omega t}$ and $e^{-j\omega t}$

Table 14. Complex exponentials on the unit circle

Function	Trajectory	Real Part	Imaginary Part
$e^{j\omega t}$	Unit circle (CCW)	$\cos(\omega t)$	$+\sin(\omega t)$
$e^{-j\omega t}$	Unit circle (CW)	$\cos(\omega t)$	$-\sin(\omega t)$

AC circuit phasors: $V(t) = V_0 e^{j\omega t} \Rightarrow \text{Re}\{V\} = V_0 \cos(\omega t)$

11.4. Hyperbolic Functions from Euler

Table 15. Hyperbolic functions defined via Euler

Function	Euler Definition	Shape
$\cosh(x)$	$\frac{e^x + e^{-x}}{2}$	Catenary (U-shape), even
$\sinh(x)$	$\frac{e^x - e^{-x}}{2}$	Odd S-shape
$\tanh(x)$	$\frac{\sinh x}{\cosh x}$	Bounded S-shape in $(-1, 1)$

Key identity: $\cosh^2(x) - \sinh^2(x) = 1$ (compare: $\cos^2 \theta + \sin^2 \theta = 1$)

Transmission line equations (power systems):

$$V(x) = V^+ \cosh(\gamma x) + V^- \sinh(\gamma x) \quad (23)$$

$$I(x) = \frac{V^+}{Z_c} \cosh(\gamma x) - \frac{V^-}{Z_c} \sinh(\gamma x) \quad (24)$$

where $\gamma = \sqrt{ZY}$ is the propagation constant and $Z_c = \sqrt{Z/Y}$ is the characteristic impedance.

11.5. Damped Oscillation: The Master Engineering Exponential

$$f(t) = e^{-\alpha t} \cdot e^{j\omega_d t} = e^{-\alpha t} [\cos(\omega_d t) + j \sin(\omega_d t)] \quad (25)$$

Real measurable part:

$$\text{Re}\{f(t)\} = e^{-\alpha t} \cos(\omega_d t) \quad (26)$$

RLC characteristic equation:

$$s^2 + 2\alpha s + \omega_n^2 = 0 \implies s = -\alpha \pm j\omega_d \quad (27)$$

where $\alpha = R/2L$, $\omega_n = 1/\sqrt{LC}$, $\omega_d = \sqrt{\omega_n^2 - \alpha^2}$.

Table 16. Three cases of damping in RLC circuits

Case	Condition	System Response
Underdamped	$\alpha < \omega_n$	Oscillatory decay: $e^{-\alpha t} \cos \omega_d t$
Critically damped	$\alpha = \omega_n$	Fastest non-oscillatory decay
Overdamped	$\alpha > \omega_n$	Slow exponential decay, no oscillation

11.6. Figure

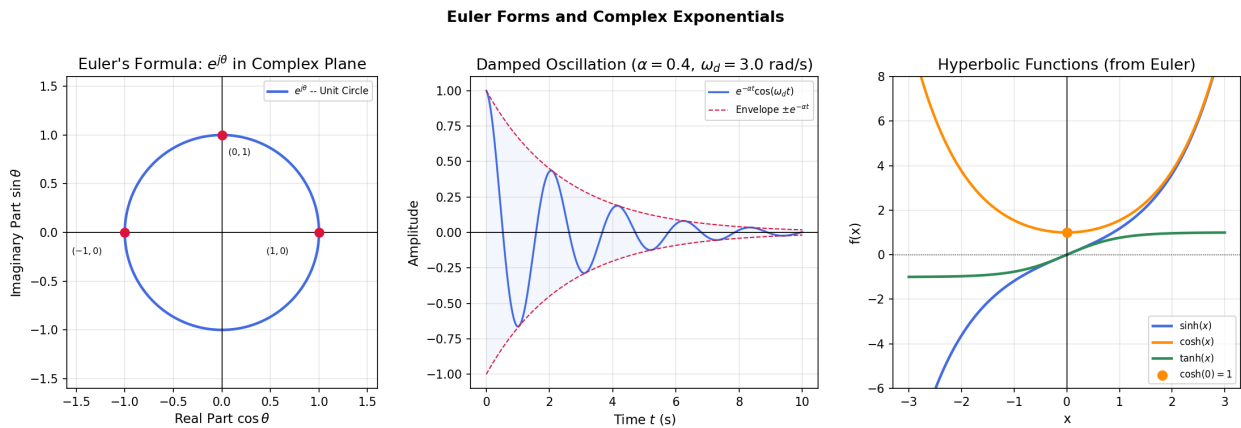


Figure 10. Euler Forms and Complex Exponentials (three panels). **Left:** Unit circle $e^{j\theta}$ in the complex plane. **Centre:** Damped oscillation $e^{-\alpha t} \cos(\omega_d t)$ with exponential envelope. **Right:** Hyperbolic functions $\sinh$, $\cosh$, $\tanh$.

12. Master Summary Tables

12.1. Table 1: Real Exponential Functions – Quick Reference

Table 17. All exponential family functions at a glance

Function	Domain	Range	HA (left)	HA (right)	VA	Intercept
e^x	$\mathbb{R}$	$(0, \infty)$	$y = 0$	None	None	$(0, 1)$
e^{-x}	$\mathbb{R}$	$(0, \infty)$	None	$y = 0$	None	$(0, 1)$
$1 - e^{-x}$	$\mathbb{R}$	$(-\infty, 1)$	None	$y = 1$	None	$(0, 0)$
$1 - e^x$	$\mathbb{R}$	$(-\infty, 1)$	$y = 1$	None	None	$(0, 0)$
$-e^{-x}$	$\mathbb{R}$	$(-\infty, 0)$	None	$y = 0^-$	None	$(0, -1)$
$\frac{1}{1 + e^{-x}}$	$\mathbb{R}$	$(0, 1)$	$y = 0$	$y = 1$	None	$(0, 0.5)$
$\tanh(x)$	$\mathbb{R}$	$(-1, 1)$	$y = -1$	$y = 1$	None	$(0, 0)$
$e^x + e^{-x}$	$\mathbb{R}$	$[2, \infty)$	None	None	None	$(0, 2)$ min
$\frac{1}{1 - e^{-x}}$	$\mathbb{R} \setminus \{0\}$	$(-\infty, 0) \cup (1, \infty)$	$y = 0^-$	$y = 1^+$	$x = 0$	None

12.2. Table 2: Engineering Applications Summary

Table 18. Exponential functions and their engineering domains

Function	Application Area	Specific Example
e^x	Thermal / Positive Feedback	BJT thermal runaway, chain reaction
e^{-x}	RC/RL Decay	Capacitor discharging, inductor decay
$1 - e^{-x}$	First-Order Step Response	RC charging, motor current buildup
$1 - e^x$	Magnetic / Fail-Safe Systems	Electromagnetic brake force drop
$-e^{-x}$	Inverting Amplifiers	Op-amp integrator negative output
Sigmoid	Neural Networks / Soft-Start	Load forecasting, motor drive ramp
$\tanh(x)$	Transmission Lines / LSTM	Voltage profile, time-series ML
$e^x + e^{-x}$	Civil / Power Engineering	Power line sag, cable catenary
$1/(1 - e^{-x})$	Resonance / Instability	Spring-mass resonance catastrophe
$e^{j\omega t}$	AC Circuits	Phasor representation, FFT
$e^{-\alpha t} \cos \omega_d t$	RLC / Vibrations	Shock absorber, seismograph ring

12.3. Table 3: Euler Forms and Their Meanings

Table 19. Complex exponential forms and physical interpretation

Form	Physical Meaning	Engineering Appearance
$e^{j\omega t}$	Pure CCW rotation (unit circle)	Steady-state AC analysis
$e^{-j\omega t}$	Pure CW rotation (unit circle)	Negative frequencies in Fourier
$e^{-\alpha t} e^{j\omega t}$	Inward spiral (stable decay)	RLC transients, damped modes
$e^{+\alpha t} e^{j\omega t}$	Outward spiral (instability)	Destructive resonance
$\cosh(x)$	U-shape, hanging cable	Power line sag, catenary
$\sinh(x)$	Odd S-curve	Transmission line current profile
$\tanh(x)$	Bounded S-shape	LSTM activation, load saturation

13. The Physical Significance of Asymptotes

Core Philosophy

Asymptotes tell you the **limits of a physical system** – what it cannot exceed, what it approaches, and where it breaks.

13.1. Horizontal Asymptotes (HA): The Steady-State / Saturation Limit

A horizontal asymptote $y = c$ represents the **steady-state value** or **saturation limit** – what the system settles to after all transients die away.

Table 20. Physical meaning of horizontal asymptotes in engineering systems

Physical System	Model	HA	Physical Meaning
Capacitor charging	$V_0(1 - e^{-t/RC})$	$V = V_0$	Cannot exceed supply voltage
Capacitor discharging	$V_0 e^{-t/RC}$	$V = 0$	Fully discharges to zero
Oven temperature	$T_{\text{set}} + (T_0 - T_{\text{set}})e^{-t/\tau}$	$T = T_{\text{set}}$	Reaches thermostat setpoint
Motor speed (sigmoid)	$\omega_{\text{max}}/(1 + e^{-kt})$	$\omega = \omega_{\text{max}}$	Cannot overspeed
Logistic population	$K/(1 + e^{-rt})$	$P = K$	Carrying capacity limit

Classroom Analogy

You step into a hot shower. The water starts cold, heats rapidly, then slows as it approaches the heater setting. Getting closer and closer to 40°C but theoretically never quite reaching it in finite time – the heater's temperature setting is the horizontal asymptote.

13.2. Vertical Asymptotes (VA): The Breakdown / Resonance Point

A vertical asymptote at $x = a$ marks a **critical failure point** or **singularity** where the system's response grows without bound.

Table 21. Physical meaning of vertical asymptotes in engineering systems

Physical System	VA Location	Physical Meaning
Spring-mass resonance	$\omega = \omega_n$	Amplitude $\rightarrow \infty$; structure fails
LC parallel resonance	$\omega = 1/\sqrt{LC}$	Impedance $\rightarrow \infty$; circuit breaks
Power system voltage collapse	$P = P_{\text{critical}}$	Voltage nose-point; blackout
Op-amp at instability	$f = f_{\text{GBW}}$	Gain $\rightarrow \infty$; oscillation begins

Classroom Analogy

Pushing a child on a swing exactly at the natural frequency – each push adds energy perfectly. Without friction, the swing would eventually reach infinite amplitude. The natural frequency is the vertical asymptote of the amplitude response.

13.3. Comparison: HA vs. VA**Table 22.** Horizontal vs. vertical asymptote – key distinctions

Feature	Horizontal Asymptote (HA)	Vertical Asymptote (VA)
Represents	Steady-state / saturation limit	Critical failure / singularity
Physical meaning	System reaches equilibrium	System resonates or breaks down
Occurrence	As $x \rightarrow \pm\infty$	At a specific finite value $x = a$
Function behaviour	Approaches finite value c smoothly	Blows up to $\pm\infty$
Engineering action	Design to reach this	Design to AVOID this
Analogy	Speed limit on a highway	A cliff edge

14. Time Constants: The Universal Engineering Concept

Definition of Time Constant τ

τ is the time for the transient to decay to $1/e \approx 36.8\%$ of its initial value, or equivalently, for the response to rise to 63.2% of its final value.

14.1. Time Constant for Common Systems

Table 23. Time constants in various first-order systems

System	Time Constant τ	Physical Meaning
RC circuit	$\tau = RC$	$\Omega \cdot F = s$
RL circuit	$\tau = L/R$	$H/\Omega = s$
Thermal system	$\tau = mC_p/hA$	$m = \text{mass}, C_p = \text{specific heat}$
First-order control	$\tau = 1/(\text{pole})$	Bandwidth $= 1/\tau$
Mechanical damper	$\tau = m/b$	$m = \text{mass}, b = \text{damping coefficient}$

14.2. The 5- τ Rule

Table 24. Transient completeness at multiples of τ

Multiple of τ	$e^{-t/\tau}$ remaining	% Charged $(1 - e^{-t/\tau})$
1τ	36.8%	63.2%
2τ	13.5%	86.5%
3τ	5.0%	95.0%
4τ	1.8%	98.2%
5τ	0.67%	99.3%

After 5τ , the transient is considered over ($< 1\%$ remaining). This determines the settling time $T_s \approx 5\tau$.

14.3. Frequency Domain Connection

In the Laplace domain, the first-order transfer function:

$$H(s) = \frac{1}{\tau s + 1} \quad (28)$$

has a **pole at** $s = -1/\tau$. The further the pole from the origin (small τ), the faster the response. The time constant is the inverse of the pole frequency.

15. Quick Revision Checklist

15.1. Before the Exam / Viva

- ✓ Can I write the Maclaurin series of e^x from memory?
- ✓ Can I sketch all 9 functions in Section 11, Table 1 without reference?
- ✓ Do I know the HA and VA for each function?
- ✓ Can I derive $\cos \theta$ and $\sin \theta$ from Euler's formula?
- ✓ Can I explain the $5\text{-}\tau$ rule and its practical significance?
- ✓ Can I write the RLC damped oscillation equation and identify α , ω_d ?
- ✓ Can I explain the physical meaning of HA vs. VA with one engineering example each?
- ✓ Can I write the derivative of sigmoid in terms of sigmoid itself?
- ✓ Can I explain why $\tanh$ is preferred over sigmoid in hidden layers?
- ✓ Can I write the transmission line voltage equation using $\cosh$ and $\sinh$?

15.2. Common Mistakes to Avoid

Common Mistakes

1. **Mistake 1:** Confusing $e^x + e^{-x}$ (catenary, range $[2, \infty)$) with $e^x - e^{-x} = 2 \sinh(x)$ (range $(-\infty, \infty)$).
2. **Mistake 2:** Forgetting that $1/(1 - e^{-x})$ is **DISCONTINUOUS** at $x = 0$.
3. **Mistake 3:** Treating $e^x \approx 1 + x$ as exact – it is only valid for $|x| \ll 1$.
4. **Mistake 4:** Confusing HA and VA in a viva: remember **HA = “where you end up”**, **VA = “where you break”**.
5. **Mistake 5:** Using $i = \sqrt{-1}$ (mathematician's notation) instead of $j = \sqrt{-1}$ in engineering contexts.

Nomenclature and Abbreviations

Part A: Mathematical Symbols

Symbol	Description
e	Euler's number; $e = \lim_{n \rightarrow \infty} (1 + 1/n)^n \approx 2.71828$
e^x	Natural exponential function; base e raised to the power x
e^{-x}	Negative exponential (decay) function
$1 - e^{-x}$	Saturating rise function; models charging / step response
$1 - e^x$	Saturating drop function; mirror of charging curve
$-e^{-x}$	Negative decay function; always negative, approaches zero from below
$\sigma(x)$	Standard sigmoid (logistic) function; $\sigma(x) = 1/(1 + e^{-x})$
$\sigma'(x)$	Derivative of sigmoid; $\sigma'(x) = \sigma(x)(1 - \sigma(x))$
$\tanh(x)$	Hyperbolic tangent; $\tanh(x) = (e^x - e^{-x})/(e^x + e^{-x})$
$\sinh(x)$	Hyperbolic sine; $\sinh(x) = (e^x - e^{-x})/2$
$\cosh(x)$	Hyperbolic cosine; $\cosh(x) = (e^x + e^{-x})/2$; minimum value = 1 at $x = 0$
$\operatorname{sech}(x)$	Hyperbolic secant; $\operatorname{sech}(x) = 1/\cosh(x)$; derivative of $\tanh(x)$
j	Imaginary unit in engineering notation; $j = \sqrt{-1}$
θ	General angle variable (radians)
ω	Angular frequency (rad/s); $\omega = 2\pi f$
ω_n	Natural (undamped) angular frequency of a second-order system (rad/s)
ω_d	Damped natural frequency; $\omega_d = \sqrt{\omega_n^2 - \alpha^2}$ (rad/s)
α	Damping factor / neper frequency (Np/s); $\alpha = R/2L$ for RLC
τ	Time constant of a first-order system (s)
γ	Propagation constant of a transmission line; $\gamma = \sqrt{ZY}$
Z_c	Characteristic impedance of a transmission line; $Z_c = \sqrt{Z/Y}$ (Ω)
Z	Series impedance per unit length of a transmission line (Ω/m)
Y	Shunt admittance per unit length of a transmission line (S/m)
s	Complex frequency variable in the Laplace domain; $s = \sigma + j\omega$
$n!$	Factorial of n ; $n! = n \times (n-1) \times \dots \times 1$, with $0! = 1$
K	Carrying capacity in logistic / population growth models
r	Intrinsic growth rate in the logistic ODE
k	Generic proportionality / rate constant in first-order ODEs
Q_0	Initial value of a quantity in an exponential growth/decay model
V_0	Initial voltage across a capacitor or inductor (V)
V_T	Thermal voltage of a p-n junction; $V_T = kT/q \approx 26$ mV at 300 K

Symbol	Description
I_S	Reverse saturation current of a BJT (A)
I_C	Collector current of a BJT (A)
V_{BE}	Base-emitter voltage of a BJT (V)
w_i	Weight of the i -th input in an artificial neuron
b	Bias term in an artificial neuron (also damping coefficient in mechanics)
F_0	Amplitude of the applied force in a forced spring-mass system (N)
m	Mass (kg)
a	Catenary parameter; related to cable tension and weight per unit length
d	Sag of a transmission line conductor (m)
L_{span}	Horizontal span length between two transmission towers (m)
T	Horizontal tension in a conductor (N)
w	Weight per unit length of a conductor (N/m)

Part B: Abbreviations and Acronyms

Abbrevia- tion	Full Form
AC	Alternating Current
ATC	Available Transfer Capability
BJT	Bipolar Junction Transistor
CCW	Counter-Clockwise
CW	Clockwise
DC	Direct Current
DNN	Deep Neural Network
EE	Electrical Engineering
FACTS	Flexible AC Transmission Systems
FFT	Fast Fourier Transform
GATE	Graduate Aptitude Test in Engineering
GBW	Gain-Bandwidth Product (of an op-amp)
HA	Horizontal Asymptote
IoT	Internet of Things
LC	Inductor-Capacitor (circuit)
LSTM	Long Short-Term Memory (recurrent neural network architecture)
MAKAUT	Maulana Abul Kalam Azad University of Technology
ML	Machine Learning

Abbrevia- tion	Full Form
MPPT	Maximum Power Point Tracking
ODE	Ordinary Differential Equation
Op-amp	Operational Amplifier
PLC	Programmable Logic Controller
PV	Photovoltaic
RC	Resistor-Capacitor (circuit)
RL	Resistor-Inductor (circuit)
RLC	Resistor-Inductor-Capacitor (circuit)
ReLU	Rectified Linear Unit (activation function)
UPFC	Unified Power Flow Controller
VA	Vertical Asymptote

Part C: Key Numerical Constants

Constant	Value and Significance
e	2.71828182845 ... – Euler's number; base of the natural logarithm
$1/e$	0.36788 ... $\approx 36.8\%$ – value of decay at $t = \tau$
$1 - 1/e$	0.63212 ... $\approx 63.2\%$ – value of rise at $t = \tau$
e^{-2}	0.13534 ... $\approx 13.5\%$ – value of decay at $t = 2\tau$
e^{-3}	0.04979 ... $\approx 5.0\%$ – value of decay at $t = 3\tau$
e^{-5}	0.00674 ... $\approx 0.67\%$ – value of decay at $t = 5\tau$ (settled)
$\sigma(0)$	0.5 – sigmoid output at the centre point
$\sigma'(0)$	0.25 – maximum slope of the sigmoid function
$\tanh'(0)$	1.0 – maximum slope of tanh; four times steeper than sigmoid
$\cosh(0)$	1.0 – minimum value of $\cosh(x)$, occurring at $x = 0$
$e^{j\pi}$	-1 – Euler's identity: $e^{j\pi} + 1 = 0$
V_T	≈ 26 mV at $T = 300$ K – thermal voltage of a p-n junction